PROJECT PROPOSAL

Original File Name

MT_CarDekho_Project_Proposal_Alex_DeBusk.pdf

Project Proposal

Domain and Problem Statement

Used car prices can vary significantly depending on a vehicle's features, condition, and history. Factors such as mileage, engine size, maximum power, fuel type, transmission, and previous ownership can all influence a vehicle's selling price.

This project will use the CarDekho dataset from Kaggle, which contains 4,023 records and 13 attributes. The goal is to explore the raw data, identify and address data quality issues, and prepare the dataset for future machine learning applications. Through data preprocessing, the original vehicle information will be transformed into a cleaner and more organized dataset suitable for analysis and model training.

Data Preprocessing Workflow

The dataset will first be divided into training and testing sets using a 70/30 split. Exploratory Data Analysis (EDA) and preprocessing will be performed using only the training data, while the testing data will remain untouched to prevent data leakage.

Missing values in numerical columns, such as mileage, engine, max power, and seats, will be replaced with their respective median

values calculated from the training data. The owner column will be converted into numerical values using ordinal encoding, while one-hot encoding will be applied to fuel, seller type, and transmission. Since selling prices are highly right-skewed, a logarithmic transformation will be applied to make their distribution more balanced. Finally, StandardScaler will be used to scale continuous numerical features to a similar range.

Final Machine Learning Objective

Once the dataset has been cleaned and preprocessed, it will be used to train and evaluate several regression models to predict used car selling prices. These models will include Linear Regression, Ridge Regression, Random Forest Regressor, and LightGBM.

The models will use the available vehicle features to estimate selling prices. Their performance will be compared using Root Mean Square Error (RMSE), which measures the difference between predicted and actual prices. A lower RMSE indicates that a model's predictions are closer to the actual selling prices.

PROJECT PROPOSAL

Original File Name

MT_ProjectProposal_Shareefah_Adeola_ITAI1371.pdf

Dataset Description and Project Proposal

Student: Shareefah Adeola

Team: ML_1371_14274 4_ITAI1371

Date: 10/03/2026

Project Title: Used Car Selling Price Prediction Using Machine Learning

Dataset Description

The selected dataset is a used-car dataset containing information about vehicles, including vehicle type, make, year, selling price, kilometers driven, fuel type, seller type, transmission, ownership history, mileage, engine size, maximum power, and number of seats. The dataset contains more than 2,000 observations and provides both numerical and categorical variables suitable for exploratory data analysis and machine-learning preprocessing.

The primary target variable for this project is **selling_price**. The objective is to investigate the factors associated with used-car selling prices and prepare the dataset for machine-learning models that can predict the selling price of a vehicle.

Data Preprocessing

The dataset was divided programmatically into **70% training data and 30% testing data using Python**. Exploratory data analysis and preprocessing were performed on the training data while keeping the testing data separate.

The preprocessing includes:

- Removing duplicate observations.
- Identifying and treating missing values using appropriate numerical median values.
- Identifying extreme values using the IQR method.
- Capping extreme km_driven values at the calculated upper IQR boundary of **188,750 kilometers**.

- Creating a new feature, **vehicle_age**, from the vehicle year using 2020 as the reference year.
- Applying one-hot encoding to categorical variables.
- Standardizing numerical predictor variables using StandardScaler.
- Using Matplotlib to visualize distributions before and after preprocessing.

Project Objective

The objective of this project is to prepare a clean and machine-learning-ready dataset that can be used to develop models for predicting used-car selling prices. The analysis will examine relationships between vehicle characteristics and selling price and will provide a foundation for training and comparing several machine-learning models.

The final processed training dataset contains **2,355 observations and 14 columns**, with no missing values or duplicate rows. The machine-learning training feature matrix contains **992 predictor features**, while the testing feature matrix contains the same 992 features.

The completed project will demonstrate the complete data-science workflow from dataset selection and exploratory analysis through data cleaning, preprocessing, feature engineering, and preparation for machine-learning modeling.

PROJECT PROPOSAL

Original File Name

MT_Project_Proposal_Yilin_Leng_ITAI1371.pdf

Project Proposal

Domain and Problem Statement

Used car prices can vary a lot depending on the condition and features of a vehicle. Factors such as mileage, engine size, power, fuel type, transmission, and ownership history can all affect the final selling price. In this project, I will work with the CarDekho dataset from Kaggle, which contains 4,023 records and 13 attributes. The purpose of this project is to examine the raw data, find and handle data quality issues, and prepare the dataset for future machine learning models. By the end of the preprocessing process, the original vehicle information will be converted into a cleaner and more useful dataset.

Data Preprocessing Workflow

First, the data will be divided into a training set and a testing set using a 70/30 split. The testing data will not be used during the EDA or the preprocessing steps to reduce the risk of data leakage. Missing values in numerical columns, including mileage, engine, Maxpower, and seats, will be filled using the median values from the training data. For categorical variables, the owner column will be changed into numerical values using ordinal encoding. One hot encoding will be used for fuel, seller type, and transmission. The selling price column will also be log transformed because its values are highly right skewed. Finally,

StandardScaler will be used to put the continuous numerical features on a similar scale.

Final Machine Learning Objective

After the data has been cleaned and prepared, the training dataset will be used to build several regression models. The models will include Linear Regression, Ridge Regression, Random Forest Regressor, and LightGBM. These models will use the available vehicle information to estimate used car selling prices. Model performance will be compared using Root Mean Square Error (RMSE). A lower RMSE means that the predicted prices are closer to the actual prices.